MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER PRACTICAL EXAMINATION
(2024 ADMISSION ONWARDS)

MG3DSCCSC202- PYTHON FOR DATA ANALYTICS

Duration of the Examination: 1.5 Hrs. Maximum Marks: 35

List of Questions

1. Create a 3×3 NumPy array containing numbers from 1 to 9. Access and print the element located in the 2nd row and 3rd column. Slice the array to print the first two rows. Iterate through all the elements of the array and print each one. Finally, use a condition to print all elements that are greater than 5.

   [A][CO3]

2. Create a 2×6 NumPy array with numbers from 1 to 12 and then reshape it into a 3×4 array. Transpose the reshaped array to interchange its rows and columns. Concatenate the transposed array with itself along axis 1 to form a larger array. Finally, modify the array by replacing all elements that are less than 7 with 0.

   [A][CO3]

3. Create two arrays a = np.array([1, 2, 3]) and b = np.array([4, 5, 6]). Perform concatenation of the arrays, vertical stacking using vstack, horizontal stacking using hstack, and finally create a NumPy array of 10 elements and split it into 2 equal parts.

   [A][CO3]

4. Create a DataFrame from the following CSV file:

Name	Age	Marks	City
Alice	20	85	New York
Bob	21	90	London
Carol	19	78	Paris
Dave	22	92	Tokyo

   Read the CSV file into a new DataFrame. Display the contents of the new DataFrame.

   [A][CO3]

5. Create a DataFrame from students.csv file containing fields Name, Age, Mark and City. Select all rows where Age is greater than 20 using the loc() function. Also, extract the first two rows using the iloc() function. Replace any missing values in the Marks column with 0 and display the updated DataFrame.

   [A][CO3]

6. Create a DataFrame from students.csv containing fields Name, Age, Mark and City. Perform descriptive statistical analysis on the Marks column. Calculate and display the count, sum, mean, median, mode, standard deviation, minimum, maximum, and cumulative sum of the Marks. [A][CO3]

7. Read a CSV file named students.csv containing fields Name, Age, Mark and City into a DataFrame. Display the first 5 rows and check for any missing values in the DataFrame. [A][CO3]

8. Create a DataFrame from students.csv containing fields Name, Age, Mark and City. Add a new column called Pass/Fail. Give 'Pass' if the Marks are 50 or more, otherwise give 'Fail'. Remove the City column from the DataFrame. Finally, display the updated DataFrame. [A][CO3]

9. Create a new DataFrame1 from student1.csv and DataFrame2 from student2.csv with at least 2 student records. The student1.csv and student2.csv contains the fields Name, Age, Mark and City. Concatenate the DataFrame1 with the DataFrame2. Finally, display the concatenated DataFrame. [A][CO3]

10. Create a DataFrame with student names and their marks. Draw a bar plot to show each student's marks and a line plot to show the trend of marks across the students. [A][CO3]

11. Create a DataFrame with employee names, their monthly sales, and their performance, such as Excellent/Good/Average. Use the sales column to draw a histogram showing the distribution of sales. Create a pie chart to show the percentage of employees in each performance category. Finally, draw a box plot to visualize the spread of sales and identify any outliers. [A][CO3]

12. Create a DataFrame containing student names, Age, and Marks. Using this DataFrame, draw a density plot for the Marks column to show its probability distribution. Then, create a scatter plot comparing Age and Marks to analyze the relationship between students' ages and their marks. Display both plots. [A][CO3]